Order of Operations

Parentheses first. Then × and ÷. Then + and □.

1. ( )
Parentheses first

2. × ÷
Times and divide, left to right

3. + −
Plus and minus, left to right

(8 + 4) × 3

$(8 + 4) \times 3 = 12 \times 3 = 36$

The parentheses go first: 8 + 4 = 12.

8 + 4 × 3

$8 + 4 \times 3 = 8 + 12 = 20$

No parentheses: times goes first.

Words to numbers

"Add 5 and 7, then multiply by 2"

$(5 + 7) \times 2 = 24$

The part you do first goes in parentheses.

Your turn!

$(10 - 4) \times 5 =$ **30**

For teachers

Goal: work out expressions with parentheses, and × and ÷ before + and − (5.OA.1–2). Say "What do we do first?" and have kids underline that part before they work it out.

Order of Operations: Practice

Parentheses first. Then × and ÷. Then + and −.

Part 1 Parentheses first Do the parentheses, then the rest.

1) $(3 + 4) \times 2 = \boxed{7} \times 2 = \boxed{14}$

2) $(15 - 6) \times 4 = \boxed{9} \times 4 = \boxed{36}$

3) $36 \div (2 + 7) = 36 \div \boxed{9} = \boxed{4}$

4) $(9 + 11) \div 5 = \boxed{20} \div 5 = \boxed{4}$

5) $6 \times (13 - 5) = 6 \times \boxed{8} = \boxed{48}$

6) $(24 - 3) \div 7 = \boxed{21} \div 7 = \boxed{3}$

7) $50 - (8 + 12) = 50 - \boxed{20} = \boxed{30}$

8) $(7 + 5) \times 8 = \boxed{12} \times 8 = \boxed{96}$

Part 2 Parentheses No help this time.

9) $(9 - 4) \times 6 = \boxed{30}$

10) $40 \div (2 + 6) = \boxed{5}$

11) $(15 + 9) \div 4 = \boxed{6}$

Order of Operations: Practice (continued)

Part 2 Parentheses (continued)

12) $7 \times (8 - 3) = 35$

13) $(6 + 6) \times 5 = 60$

14) $(30 - 12) \div 3 = 6$

15) $9 \times (2 + 7) = 81$

16) $(45 - 5) \div 8 = 5$

17) $100 - (25 + 35) = 40$

18) $(14 + 6) \times 3 = 60$

19) $72 \div (12 - 4) = 9$

20) $(11 - 2) \times 9 = 81$

Part 3 Times and divide first Do × or ÷, then the rest.

21) $5 + 3 \times 4 = 5 + 12 = 17$

22) $20 - 12 \div 3 = 20 - 4 = 16$

Order of Operations: Practice (continued)

Part 3 Times and divide first (continued)

23) $6 \times 7 + 8 = \boxed{42} + 8 = \boxed{50}$

24) $30 \div 5 - 2 = \boxed{6} - 2 = \boxed{4}$

25) $9 + 8 \times 2 = 9 + \boxed{16} = \boxed{25}$

26) $40 - 6 \times 5 = 40 - \boxed{30} = \boxed{10}$

27) $48 \div 6 + 15 = \boxed{8} + 15 = \boxed{23}$

28) $17 + 36 \div 4 = 17 + \boxed{9} = \boxed{26}$

Part 4 Times and divide first No help this time.

29) $4 + 6 \times 3 = \boxed{22}$

30) $50 - 8 \times 4 = \boxed{18}$

31) $7 \times 6 - 12 = \boxed{30}$

32) $45 \div 9 + 11 = \boxed{16}$

33) $3 + 64 \div 8 = \boxed{11}$

34) $9 \times 9 - 30 = \boxed{51}$

Order of Operations: Practice (continued)

Part 4 Times and divide first (continued)

35) $100 - 56 \div 7 =$ 92

36) $8 + 7 \times 7 =$ 57

37) $63 \div 7 + 26 =$ 35

38) $25 - 4 \times 5 =$ 5

39) $12 \times 3 + 14 =$ 50

40) $60 - 42 \div 6 =$ 53

Part 5 Same numbers, different answers No help this time. Parentheses change the order.

41) $(2 + 3) \times 4 =$ 20 and $2 + 3 \times 4 =$ 14

42) $(10 - 2) \times 3 =$ 24 and $10 - 2 \times 3 =$ 4

43) $(18 + 6) \div 6 =$ 4 and $18 + 6 \div 6 =$ 19

44) $5 \times (4 + 2) =$ 30 and $5 \times 4 + 2 =$ 22

Order of Operations: Practice (continued)

Part 6 Words to numbers No help this time. Write the expression, then work it out.

45) Add 6 and 9, then multiply by 3: 45

46) Take 8 from 20, then divide by 4: 3

47) Multiply 7 by 5, then add 12: 47

48) Add 14 and 22, then divide by 6: 6

49) Divide 81 by 9, then take away 4: 5

50) Take 3 from 11, then multiply by 7: 56

Part 7 How many times as big? No help this time. Don't work it out.

51) $4 \times (27 + 35)$ is 4 times as big as $27 + 35$

52) $6 \times (189 + 52)$ is 6 times as big as $189 + 52$

53) $3 \times (418 + 296)$ is 3 times as big as $418 + 296$

54) $8 \times (75 + 64)$ is 8 times as big as $75 + 64$

Order of Operations: Practice (continued)

Part 8 Parentheses first

55) $(3 + 4) \times 6 =$ 42

56) $(12 + 3) \times 2 =$ 30

57) $(11 + 4) \times 3 =$ 45

58) $(9 + 8) \times 7 =$ 119

59) $(11 + 8) \times 4 =$ 76

60) $(7 + 9) \times 9 =$ 144

61) $(8 + 11) \times 7 =$ 133

62) $(5 + 10) \times 5 =$ 75

63) $(5 + 14) \times 3 =$ 57

64) $(15 + 4) \times 4 =$ 76

65) $(4 + 11) \times 8 =$ 120

66) $(10 + 11) \times 2 =$ 42

Order of Operations: Practice (continued)

Part 8 Parentheses first (continued)

67) $(3 + 12) \times 6 = 90$

68) $(7 + 9) \times 2 = 32$

69) $(12 + 12) \times 9 = 216$

70) $(10 + 8) \times 7 = 126$

71) $(5 + 4) \times 7 = 63$

72) $(7 + 8) \times 8 = 120$

Part 9 Times first

73) $40 + 3 \times 7 = 61$

74) $22 + 6 \times 7 = 64$

75) $3 + 6 \times 8 = 51$

76) $36 + 3 \times 2 = 42$

77) $22 + 5 \times 6 = 52$

78) $3 + 8 \times 8 = 67$

Order of Operations: Practice (continued)

Part 9 Times first (continued)

79) $6 + 5 \times 4 =$ 26

80) $40 + 6 \times 7 =$ 82

81) $22 + 4 \times 3 =$ 34

82) $17 + 8 \times 7 =$ 73

83) $20 + 9 \times 9 =$ 101

84) $34 + 6 \times 9 =$ 88

85) $14 + 7 \times 5 =$ 49

86) $33 + 5 \times 3 =$ 48

87) $39 + 3 \times 8 =$ 63

88) $32 + 5 \times 7 =$ 67

89) $8 + 3 \times 3 =$ 17

90) $9 + 4 \times 8 =$ 41

Order of Operations: Practice (continued)

Part 10 Divide first

91) $21 - 27 \div 3 = 12$

92) $40 - 36 \div 4 = 31$

93) $59 - 21 \div 7 = 56$

94) $40 - 40 \div 5 = 32$

95) $54 - 6 \div 3 = 52$

96) $26 - 72 \div 8 = 17$

97) $35 - 56 \div 8 = 28$

98) $20 - 40 \div 5 = 12$

99) $59 - 8 \div 4 = 57$

100) $29 - 32 \div 4 = 21$

101) $23 - 15 \div 3 = 18$

102) $35 - 45 \div 9 = 30$

Order of Operations: Practice (continued)

Part 11 Remember? Problems from earlier packets.

103) $\frac{1}{8} \div 3 = \frac{1}{24}$

104) $5 \div \frac{1}{3} = 15$

105) $6 \div 8 = \frac{6}{8}$

106) $7 \times 0.9 = 6.3$

107) 5 cm × 2 cm × 6 cm = 60 cubic cm

108) 3.6 to the nearest whole number = 4

Part 12 Word problems Write one expression, then work it out.

109) Mia buys 3 packs of 8 pens and 5 more pens. How many pens does she have?

Number sentence 3 × 8 + 5 = 29

Answer 29 pens

Draw or show your work

110) Leo has 40 stickers. He gives away 12 and shares the rest among 4 friends. How many each?

Number sentence (40 − 12) ÷ 4 = 7

Answer 7 stickers

Draw or show your work

111) A ticket costs $6. 4 adults and 5 kids go. How much do they pay in all?

Number sentence (4 + 5) × 6 = 54

Answer $54

Draw or show your work

Order of Operations: Practice (continued)

Part 12 Word problems (continued)

112) Zoe has 50 beads. She uses 6 beads on each of 7 bracelets. How many beads are left?

Number sentence $50 - 6 \times 7 = 8$

Answer 8 beads

Draw or show your work

113) Challenge: Ben reads 15 pages a day for 4 days, then 25 pages. Sam reads 80. Who reads more?

Number sentence $15 \times 4 + 25 = 85 > 80$

Answer Ben

Draw or show your work

Check-In: Order of Operations

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Parentheses

1) $(20 - 11) \times 6 =$ 54

2) $(64 - 16) \div 6 =$ 8

3) $(33 + 7) \div 5 =$ 8

Part 2 Times and divide first

4) $70 - 8 \times 8 =$ 6

5) $8 \times 4 - 9 =$ 23

6) $2 + 9 \times 6 =$ 56

Part 3 Words to numbers

7) Take 6 from 30, then divide by 8: 3

8) Multiply 8 by 6, then take away 10: 38